

AVCDL Standards Cross Reference

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Abstract

This document provides a summary the standards and regulations addressed by each **AVCDL** phase requirement.

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Overview

The **AVCDL** ^[1] is a cybersecurity development lifecycle intended to support safety-critical, cyber-physical system development, as well as certification under various standards and regulations in those domains. The standards and regulations included in this document are:

- **ISO/SAE 21434** ^[2] work products
- **ISO 24089** ^[5] work products
- **UN R155** ^[3] (requirements and supplemental sections)
- **UN R156** ^[6] requirements
- **CMMC** ^[4] practices

This document provides a concise list of the elements of various standards and regulations that each AVCDL phase requirements provides support for.

Note: It is not the intent of this document to explore either the **AVCDL** phase requirements or supported elements of the various standards and regulations in detail, merely to provide a summary. Much more extensive material is provide within the various **AVCDL** documents.

Note: A more compressed form of this information can be found in the **AVCDL mappings** workbook ^[5].

Note: The **UN R155** lists are divided into requirements and supplemental information. The supplemental list is not strictly necessary for compliance but is supported by various **AVCDL** processes.

Organization

Each **AVCDL** phase will be addressed in turn providing a list of the work products for the individual phase requirements.

Foundation Phase

AVCDL-Foundation-1

Training

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.2.2.2(a)] cybersecurity management

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.1(b)] production phase CSMS

[7.2.2.1(c)] post-production CSMS

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	AT.2.056, AT.2.057
3	none
4	none
5	none

AVCDL-Foundation-2

Roles and Responsibilities

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

[WP-5.4.1] Software update project plan

[WP-9.4.1] Software update campaign plan documentation

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.1(b)] production phase CSMS

[7.2.2.1(c)] post-production CSMS

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Foundation-3

Toolchain Support

ISO/SAE 21434 Required Work Products

[WP-05-04] Evidence of tool management

[WP-10-03] Documentation of the modeling, design, or programming languages and coding guidelines

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(h)] cybersecurity controls tracking

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	CM.2.061, CM.2.064, CM.2.065, CM.2.066, MA.2.112
3	AU.3.049, AU.3.052, CM.3.068, MA.3.116, CA.3.162
4	CM.4.073
5	AU.5.055, CM.5.074, RM.5.152

AVCDL-Foundation-4

Definition of Security Requirements

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(g)] adaptable monitoring / response

[7.3.4] type risk protection

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	CA.2.157
3	none
4	none
5	none

AVCDL-Foundation-5

Protect the Code

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	AC.1.001, AC.1.002, AC.1.003, AC.1.004, IA.1.076, IA.1.077, PE.1.131, PE.1.133, PE.1.134
2	AC.2.007, AC.2.008, AC.2.009, AC.2.010, IA.2.081, IA.2.082
3	AC.3.017, AC.3.018, AC.3.019, AC.3.014, AM.3.036, IA.3.083, IA.3.084, IS.3.086
4	AM.4.226
5	none

AVCDL-Foundation-6

Ensure Release Integrity

ISO/SAE 21434 Required Work Products

[WP-12-01] Production control plan

ISO 24089 Required Work Products

- [WP-5.4.5] Documentation of integrity preservation process
- [WP-6.4.4] Software update package documentation
- [WP-7.4.4] Software update package documentation
- [WP-8.4.1] Documentation of update package contents and targets
- [WP-8.4.2] Minimum required software update package
- [WP-9.4.1] Software update campaign plan documentation

UN R155 Requirements

- [7.2.2.1(b)] production phase CSMS
- [7.2.2.2(g)] adaptable monitoring / response

UN R155 Supplemental

- [7.2.2.1(a)] development phase CSMS

UN R156 Requirements

- [7.1.2.5(g)] update execution mechanism documentation
- [7.1.2.5(h)] update confirmation documentation
- [7.1.3.1] update payload integrity assurance
- [7.2.1.1] update authenticity / integrity
- [7.2.2.1.1] reversion to good state on update failure
- [7.2.2.1.2] sufficient power to perform update
- [7.2.2.1.3] safe state to perform update
- [7.2.2.3] no updates putting vehicle in unsafe state while driving
- [7.2.2.5] no updates until all precautions met

CMMC Applicable Practices

Level	Practice
1	AC.1.001, AC.1.002, AC.1.003, AC.1.004, PE.1.131, PE.1.133, PE.1.134
2	AC.2.007, AC.2.008, AC.2.009, AC.2.010
3	AC.3.017, AC.3.018, AC.3.019, AC.3.014, AM.3.036
4	AM.4.226
5	none

AVCDL-Foundation-7

Incident Response Plan

ISO/SAE 21434 Required Work Products

- [WP-07-01] Sources for cybersecurity monitoring
- [WP-07-02] Triage triggers of cybersecurity information
- [WP-08-03] Cybersecurity event triage
- [WP-08-04] Cybersecurity event assessment
- [WP-08-05] Vulnerability analysis
- [WP-08-06] Evidence of managed vulnerabilities
- [WP-08-X1] Apply incident response protocols
- [WP-13-01] Cybersecurity incident response plan

ISO 24089 Required Work Products

- [WP-4.4.4] Information sharing policy
- [WP-6.4.1] Cybersecurity risk management documentation
- [WP-7.4.1] Risk management documentation

UN R155 Requirements

- [7.2.2.1(c)] post-production phase CSMS
- [7.2.2.2(f)] risk assessment kept current
- [7.2.2.2(g)] adaptable monitoring / response
- [7.2.2.2(h)] cybersecurity controls tracking
- [7.2.2.3] timely risk mitigation
- [7.2.2.4(b)] threat extraction from vehicle logs
- [7.3.7(b)] vehicle cybersecurity monitoring
- [7.3.7(c)] provide forensic capability
- [7.4.1] periodic monitoring report

UN R155 Supplemental

[7.2.2.1(b)] production phase CSMS

[7.3.2] management of type

UN R156 Requirements

[7.1.1.12] update information regulatory disclosure

[7.1.3.2] secure update processes

CMMC Applicable Practices

Level	Practice
1	none
2	IR.2.092
3	none
4	IR.4.100
5	IR.5.106

AVCDL-Foundation-8

Decommissioning Plan

ISO/SAE 21434 Required Work Products

[WP-14-01] Procedures to communicate end of cybersecurity support

[WP-14-X1] Decommissioning requirements

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.2.2.1(c)] post-production CSMS

UN R155 Supplemental

none

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AM.3.036
4	AM.4.226
5	none

AVCDL-Foundation-9

Threat Prioritization Plan

ISO/SAE 21434 Required Work Products

[WP-08-06] Evidence of managed vulnerabilities

ISO 24089 Required Work Products

[WP-6.4.1] Cybersecurity risk management documentation

[WP-7.4.1] Risk management documentation

UN R155 Requirements

[7.2.2.1(c)] post-production CSMS

[7.2.2.2(f)] risk assessment kept current

[7.2.2.2(g)] adaptable monitoring / response

[7.2.2.3] timely risk mitigation

[7.4.1] periodic monitoring report

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(c)] risk assessment / treatment

[7.3.2] management of type

UN R156 Requirements

[7.1.3.2] secure update processes

CMMC Applicable Practices

Level	Practice
1	none
2	CA.2.159
3	RM.3.146
4	none
5	none

AVCDL-Foundation-10

Deployment Plan

ISO/SAE 21434 Required Work Products

[WP-12-01] Production control plan

[WP-13-X2] Update plan

ISO 24089 Required Work Products

[WP-5.4.4] Interoperability confirmation documentation

[WP-5.4.5] Documentation of integrity preservation process

[WP-6.4.4] Software update package documentation

[WP-7.4.4] Software update package documentation

[WP-8.4.1] Documentation of update package contents and targets

[WP-8.4.2] Minimum required software update package

[WP-9.4.1] Software update campaign plan documentation

UN R155 Requirements

[7.2.2.1(b)] production phase CSMS

[7.2.2.1(c)] post-production phase CSMS

[7.2.2.2(g)] adaptable monitoring / response

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

UN R156 Requirements

- [7.1.1.6] update target vehicle identification
- [7.1.2.2] pre/post update system configuration
- [7.1.2.4] campaign target vehicle documentation
- [7.1.2.5(b)] update affected systems / functions documentation
- [7.1.2.5(g)] update execution mechanism documentation
- [7.1.2.5(h)] update confirmation documentation
- [7.1.3.1] update payload integrity assurance
- [7.2.1.1] update authenticity / integrity
- [7.2.2.1.1] reversion to good state on update failure
- [7.2.2.1.2] sufficient power to perform update
- [7.2.2.1.3] safe state to perform update
- [7.2.2.3] no updates putting vehicle in unsafe state while driving
- [7.2.2.5] no updates until all precautions met

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AM.3.036
4	AM.4.226
5	none

Requirements Phase

AVCDL-Requirements-1

Security Requirements Definition

ISO/SAE 21434 Required Work Products

[WP-09-01] Item definition

[WP-10-02] Cybersecurity requirements for post-development

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.2.2.1(a)] development phase CSMS

UN R155 Supplemental

[7.3.4] type risk protection

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	CA.2.157
3	SC.3.177, SC.3.181, SC.3.183, SC.3.185, SC.3.186, SC.3.187, SC.3.190, SC.3.191
4	none
5	none

AVCDL-Requirements-2

Requirements Gate

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.3.6] sufficient testing

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

Design Phase

AVCDL-Design-1

Apply Security Requirements and Risk Information to Design

ISO/SAE 21434 Required Work Products

- [WP-09-06] Cybersecurity concept
- [WP-10-01] Refined cybersecurity specification
- [WP-15-02] Identified assets and cybersecurity properties

ISO 24089 Required Work Products

none

UN R155 Requirements

- [7.2.2.1(a)] development phase CSMS
- [7.2.2.2(b)] risk identification
- [7.3.3] critical element risk assessment
- [7.3.4] type risk protection

UN R155 Supplemental

- [7.3.8] use standard crypto modules

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052, SC.3.180
4	none
5	AU.5.055

AVCDL-Design-2

Security Design Review

ISO/SAE 21434 Required Work Products

- [WP-06-03] Cybersecurity assessment report
- [WP-09-07] Verification report of cybersecurity concept

ISO 24089 Required Work Products

none

UN R155 Requirements

- [7.2.2.1(a)] development phase CSMS
- [7.2.2.2(d)] verification of risk management

UN R155 Supplemental

- [7.2.2.2(e)] cybersecurity testing

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055, SC.5.230

AVCDL-Design-3

Attack Surface Reduction

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(b)] risk identification

[7.2.2.2(c)] risk assessment / treatment

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	RM.2.143
3	AU.3.049, AU.3.052, RM.3.144
4	none
5	AU.5.055

AVCDL-Design-4

Threat Modeling

ISO/SAE 21434 Required Work Products

- [WP-09-02] Threat analysis and risk assessment
- [WP-09-03] Cybersecurity goals
- [WP-09-04] Cybersecurity claims
- [WP-09-05] Verification report
- [WP-15-01] Damage scenarios
- [WP-15-03] Threat scenarios
- [WP-15-04] Impact rating
- [WP-15-05] Attack paths
- [WP-15-06] Attack feasibility rating
- [WP-15-07] Risk values
- [WP-15-08] Risk treatment decision per threat scenario

ISO 24089 Required Work Products

- [WP-6.4.1] Cybersecurity risk management documentation
- [WP-7.4.1] Risk management documentation

UN R155 Requirements

- [7.2.2.1(a)] development phase CSMS
- [7.2.2.1(c)] post-production phase CSMS
- [7.2.2.2(b)] risk identification
- [7.2.2.2(c)] risk treatment
- [7.2.2.2(d)] verification of risk management
- [7.2.2.2(f)] risk assessment kept current
- [7.2.2.2.(g)] adaptable monitoring / response
- [7.2.2.3] timely risk management
- [7.3.3] critical element risk assessment
- [7.3.4] type risk protection
- [7.3.5] type risk countermeasures
- [7.4.1] periodic monitoring report

UN R155 Supplemental

- [7.2.2.1(b)] production phase CSMS
- [7.3.2] management of type

UN R156 Requirements

- [7.1.3.2] secure update processes

CMMC Applicable Practices

Level	Practice
1	none
2	RM.2.143
3	AU.3.049, AU.3.052, RM.3.144
4	none
5	AU.5.055

AVCDL-Design-5

Design Gate

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.3.6] sufficient testing

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

Implementation Phase

AVCDL-Implementation-1

Use Approved Tools

ISO/SAE 21434 Required Work Products

[WP-05-05] Evidence of tool management

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(h)] cybersecurity controls tracking

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

AVCDL-Implementation-2

Configure Build Process to Improve Security

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(h)] cybersecurity controls tracking

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	CM.2.061, CM.2.064, CM.2.065, CM.2.066
3	CM.3.068
4	CM.4.073
5	CM.5.074

AVCDL-Implementation-3

Use Secure Settings by Default

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	CM.2.061, CM.2.064, CM.2.065, CM.2.066
3	CM.3.068
4	CM.4.073
5	CM.5.074

AVCDL-Implementation-4

Reuse Well-Secured Software

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.1(b)] production phase CSMS

[7.2.2.2(h)] cybersecurity controls tracking

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Implementation-5

Code Securely

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.3.4] type risk protection

[7.3.8] use standard crypto modules

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	SC.2.179
3	MP.3.125, SC.3.177, SC.3.181, SC.3.183, SC.3.185, SC.3.186, SC.3.187, SC.3.190, SC.3.191
4	SC.4.197
5	none

AVCDL-Implementation-6

Deprecate Unsafe Functions

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(h)] cybersecurity controls tracking

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Implementation-7

Static Analysis

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Implementation-8

Dynamic Program Analysis

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	none
4	none
5	SI.5.222

AVCDL-Implementation-9

Security Code Review

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052, CA.3.162
4	none
5	AU.5.055

AVCDL-Implementation-10

Fuzz Testing

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Implementation-11

Implementation Gate

ISO/SAE 21434 Required Work Products

[WP-10-07] Integration and verification reports

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

[7.3.6] sufficient testing

UN R155 Supplemental

none

UN R156 Requirements

none

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

Verification Phase

AVCDL-Verification-1

Penetration Testing

ISO/SAE 21434 Required Work Products

[WP-10-06] Integration and Verification Specification

ISO 24089 Required Work Products

[WP-8.4.3] Verification and validation documentation

UN R155 Requirements

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] sufficient testing

UN R155 Supplemental

none

UN R156 Requirements

[7.1.2.5(i)] update V&V documentation

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	none
4	CA.4.164, CA.4.227
5	none

AVCDL-Verification-2

Threat Model Review

ISO/SAE 21434 Required Work Products

[WP-10-04] Verification report for the refined cybersecurity specification

[WP-10-05] Vulnerability analysis report

[WP-10-06] Integration and Verification Specification

ISO 24089 Required Work Products

[WP-8.4.3] Verification and validation documentation

UN R155 Requirements

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

UN R155 Supplemental

[7.2.2.2(d)] verification of risk management

[7.2.2.2(f)] risk assessment kept current

UN R156 Requirements

[7.1.2.5(i)] update V&V documentation

CMMC Applicable Practices

Level	Practice
1	none
2	RM.2.143
3	RM.3.144
4	none
5	none

AVCDL-Verification-3

Attack Surface Analysis Review

ISO/SAE 21434 Required Work Products

[WP-10-04] Verification report for the refined cybersecurity specification

[WP-10-05] Vulnerability analysis report

[WP-10-06] Integration and Verification Specification

ISO 24089 Required Work Products

[WP-8.4.3] Verification and validation documentation

UN R155 Requirements

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(e)] cybersecurity testing

UN R155 Supplemental

[7.2.2.2(d)] verification of risk management

[7.2.2.2(f)] risk assessment kept current

UN R156 Requirements

[7.1.2.5(i)] update V&V documentation

CMMC Applicable Practices

Level	Practice
1	none
2	RM.2.143
3	RM.3.144
4	none
5	none

AVCDL-Verification-4

Verification Gate

ISO/SAE 21434 Required Work Products

[WP-10-07] Integration and verification reports

[WP-11-01] Validation report

ISO 24089 Required Work Products

[WP-8.4.3] Verification and validation documentation

[WP-8.4.4] Release approval documentation

UN R155 Requirements

[7.2.2.1(a)] development phase CSMS

[7.2.2.2(d)] verification of risk management

[7.2.2.2(e)] cybersecurity testing

[7.3.6] sufficient testing

UN R155 Supplemental

none

UN R156 Requirements

[7.1.2.5(i)] update V&V documentation

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

Release Phase

AVCDL-Release-1

Final Security Review

ISO/SAE 21434 Required Work Products

[WP-06-04] Release for post-development report

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

UN R156 Requirements

[7.1.2.5(i)] update V&V documentation

CMMC Applicable Practices

Level	Practice
1	none
2	RM.2.143
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

AVCDL-Release-2

Archive

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

- [WP-5.4.5] Documentation of integrity preservation process
- [WP-6.4.4] Software update package documentation
- [WP-7.4.4] Software update package documentation
- [WP-8.4.2] Minimum required software update package
- [WP-9.4.1] Software update campaign plan documentation

UN R155 Requirements

none

UN R155 Supplemental

- [7.2.2.1(a)] development phase CSMS

UN R156 Requirements

- [7.1.1.6] update target vehicle identification
- [7.1.2.4] campaign target vehicle documentation
- [7.1.2.5(b)] update affected systems / functions documentation
- [7.1.2.5(h)] update confirmation documentation
- [7.2.1.1] update authenticity / integrity
- [7.2.2.1.1] reversion to good state on update failure
- [7.2.2.1.2] sufficient power to perform update
- [7.2.2.1.3] safe state to perform update
- [7.2.2.3] no updates putting vehicle in unsafe state while driving
- [7.2.2.5] no updates until all precautions met

CMMC Applicable Practices

Level	Practice
1	PE.1.131, PE.1.133, PE.1.134
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055, RE.5.140

AVCDL-Release-3

Release Gate

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

none

UN R155 Supplemental

[7.2.2.1(a)] development phase CSMS

[7.3.6] sufficient testing

UN R156 Requirements

[7.1.2.5(i)] update V&V documentation

CMMC Applicable Practices

Level	Practice
1	none
2	none
3	AU.3.049, AU.3.052
4	none
5	AU.5.055

Operation Phase

AVCDL-Operation-1

Identify and Confirm Vulnerabilities

ISO/SAE 21434 Required Work Products

- [WP-08-03] Cybersecurity event triage
- [WP-08-04] Cybersecurity event assessment
- [WP-08-06] Evidence of managed vulnerabilities
- [WP-08-X1] Apply incident response protocols
- [WP-13-X1] Cybersecurity incident response plan implementation

ISO 24089 Required Work Products

- [WP-6.4.1] Cybersecurity risk management documentation
- [WP-7.4.1] Risk management documentation

UN R155 Requirements

- [7.2.2.1(c)] post-production phase CSMS
- [7.2.2.2(f)] risk assessment kept current
- [7.2.2.2(g)] adaptable monitoring / response
- [7.2.2.2(h)] cybersecurity controls tracking
- [7.2.2.3] timely risk mitigation
- [7.2.2.4(b)] threat extraction from vehicle logs
- [7.4.1] periodic monitoring report

UN R155 Supplemental

- [7.2.2.1(b)] production phase CSMS
- [7.3.2] management of type

UN R156 Requirements

- [7.1.3.2] secure update processes

CMMC Applicable Practices

Level	Practice
1	SI.1.210
2	IR.2.093, IR.2.096
3	IR.3.098, SA.3.169
4	RM4.149, RM.4.150, SA.4.171, SA.4.173, SI.4.221
5	IR.5.102

AVCDL-Operation-2

Assess and Prioritize Remediation

ISO/SAE 21434 Required Work Products

- [WP-08-05] Vulnerability analysis
- [WP-08-06] Evidence of managed vulnerabilities
- [WP-08-X1] Apply incident response protocols
- [WP-13-X1] Cybersecurity incident response plan implementation

ISO 24089 Required Work Products

- [WP-6.4.1] Cybersecurity risk management documentation
- [WP-7.4.1] Risk management documentation

UN R155 Requirements

- [7.2.2.1(c)] post-production CSMS
- [7.2.2.2(f)] risk assessment kept current
- [7.2.2.2(g)] adaptable monitoring / response
- [7.2.2.3] timely risk mitigation
- [7.3.7(c)] provide forensic capability
- [7.4.1] periodic monitoring report

UN R155 Supplemental

- [7.2.2.1(b)] production phase CSMS
- [7.2.2.2(h)] cybersecurity controls tracking
- [7.3.2] management of type

UN R156 Requirements

- [7.1.3.2] secure update processes

CMMC Applicable Practices

Level	Practice
1	SI.1.210
2	IR.2.094, RM.2.143
3	none
4	none
5	none

AVCDL-Operation-3

Root Cause Vulnerabilities

ISO/SAE 21434 Required Work Products

- [WP-08-05] Vulnerability analysis
- [WP-08-06] Evidence of managed vulnerabilities
- [WP-08-X1] Apply incident response protocols
- [WP-13-X1] Cybersecurity incident response plan implementation

ISO 24089 Required Work Products

- [WP-6.4.1] Cybersecurity risk management documentation
- [WP-7.4.1] Risk management documentation

UN R155 Requirements

- [7.2.2.1(c)] post-production CSMS
- [7.2.2.2(f)] risk assessment kept current
- [7.2.2.2(g)] adaptable monitoring / response
- [7.2.2.3] timely risk mitigation
- [7.3.7(c)] provide forensic capability

UN R155 Supplemental

- [7.2.2.1(b)] production phase CSMS
- [7.2.2.2(h)] cybersecurity controls tracking
- [7.3.2] management of type

UN R156 Requirements

- [7.1.3.2] secure update processes

CMMC Applicable Practices

Level	Practice
1	SI.1.210
2	IR.2.094, RM.2.143
3	none
4	none
5	none

AVCDL-Operation-4

Secure Deployment

ISO/SAE 21434 Required Work Products

[WP-12-X1] Production control plan implementation

ISO 24089 Required Work Products

- [WP-5.4.4] Interoperability confirmation documentation
- [WP-5.4.5] Documentation of integrity preservation process
- [WP-6.4.4] Software update package documentation
- [WP-7.4.4] Software update package documentation
- [WP-8.4.1] Documentation of update package contents and targets
- [WP-8.4.2] Minimum required software update package
- [WP-9.4.1] Software update campaign plan documentation

UN R155 Requirements

[7.2.2.1(b)] production phase CSMS

UN R155 Supplemental

[7.2.2.2(h)] cybersecurity controls tracking

UN R156 Requirements

- [7.1.1.6] update target vehicle identification
- [7.1.2.2] pre/post update system configuration
- [7.1.2.4] campaign target vehicle documentation
- [7.1.2.5(b)] update affected systems / functions documentation
- [7.1.2.5(g)] update execution mechanism documentation
- [7.1.2.5(h)] update confirmation documentation
- [7.1.3.1] update payload integrity assurance
- [7.2.1.1] update authenticity / integrity
- [7.2.2.1.1] reversion to good state on update failure
- [7.2.2.1.2] sufficient power to perform update
- [7.2.2.1.3] safe state to perform update
- [7.2.2.3] no updates putting vehicle in unsafe state while driving
- [7.2.2.5] no updates until all precautions met

CMMC Applicable Practices

Level	Practice
1	AC.1.001, AC.1.002, AC.1.003, AC.1.004, IA.1.076, IA.1.077, PE.1.131, PE.1.133, PE.1.134
2	AC.2.007, AC.2.008, AC.2.009, AC.2.010, IA.2.081, IA.2.082, MP.2.119, MP.2.120, MP.2.121, SC.2.179
3	AC.3.017, AC.3.018, AC.3.019, AC.3.014, AM.3.036, AU.3.049, AU.3.052, IA.3.083, IA.3.084, IS.3.086
4	AM.4.226
5	AU.5.055

Decommissioning Phase

AVCDL-Decommissioning-1

Apply Decommissioning Protocol

ISO/SAE 21434 Required Work Products

none

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.2.2.1(c)] post-production CSMS

UN R155 Supplemental

none

UN R156 Requirements

[] cybersecurity

CMMC Applicable Practices

Level	Practice
1	AC.1.001, AC.1.002, AC.1.003, AC.1.004, IA.1.076, IA.1.077, MP.1.118, PE.1.131, PE.1.133, PE.1.134
2	AC.2.007, AC.2.008, AC.2.009, AC.2.010, IA.2.081, IA.2.082, SC.2.179
3	AC.3.017, AC.3.018, AC.3.019, AC.3.014, AM.3.036, AU.3.049, AU.3.052, IA.3.083, IA.3.084, IS.3.086, MA.3.115, PE.3.136
4	AM.4.226
5	AU.5.055

Supplier Processes

AVCDL-Supplier-1

AVCMDS

ISO/SAE 21434 Required Work Products

[WP-07-X1] Supplier capability

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.3.2] management of type

UN R155 Supplemental

[7.2.2.5] supplier deficiency management

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Supplier-2

Supplier Self-reported Maturity

ISO/SAE 21434 Required Work Products

[WP-07-X1] Supplier capability

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.3.2] management of type

UN R155 Supplemental

[7.2.2.5] supplier deficiency management

UN R156 Requirements

none

CMMC Applicable Practices

none

AVCDL-Supplier-3

Cybersecurity Interface Agreement

ISO/SAE 21434 Required Work Products

[WP-07-01] Cybersecurity interface agreement

ISO 24089 Required Work Products

none

UN R155 Requirements

[7.2.2.2(f)] risk assessment kept current

[7.2.2.2(g)] adaptable monitoring / response

[7.2.2.5] supplier deficiency management

[7.3.2] management of type

UN R155 Supplemental

none

UN R156 Requirements

none

CMMC Applicable Practices

none

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